INCH-POUND

MIL-PRF-39015/3M
w/ Amendment 1
9 June 2017
SUPERSEDING
MIL-PRF-39015/3M
5 October 2006

PERFORMANCE SPECIFICATION SHEET

# RESISTOR, VARIABLE, WIREWOUND (LEAD-SCREW ACTUATED), NONESTABLISHED RELIABILITY AND ESTABLISHED RELIABILITY STYLE RTR24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39015.

## 1. SCOPE

1.1 Scope. This specification covers the associated requirements for style RTR24, lead-screw actuated, wire-wound, variable resistors. The resistors are available in a ±5 percent resistance tolerance and terminal types L, P, W, and X.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the document number and slash sheet, resistance value designator, terminal type, and product level designator. The PIN is derived in accordance with MIL-PRF-39015 and in the following format.

| M39015/3 | - | 001 | P | M |
|---|---|---|---|---|
| Associated specification number | | Resistance value designator | Terminals | Product level designator |

NOTE: The slash "/" and the dash "-" are needed in the procurement of this part.

Comments, suggestions, or questions on this document should be emailed to usarmy.apg.cerdec.mbx.standardization-crx@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5905

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:30Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document user are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-39015 - Resistor, Variable, Wirewound, (Lead Screw Actuated), Nonestablished Reliability and Established Reliability, General Specification for

* (Copies of these documents are available online at http://quicksearch.dla.mil/.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-39015.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be 3/4 watt, based on full load operation at an ambient temperature of 85°C.

3.4 Nominal resistance value, maximum resolution, and maximum rated ac or dc working voltage. Nominal resistance value, maximum resolution, and maximum rated ac or dc working voltages shall be as specified in table I.

3.5 Actual effective electrical travel. Actual effective electrical travel shall be 15 turns minimum, and 30 turns maximum.

3.6 Operating torque. Operating torque shall be a maximum of 5 ounce-inches.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:30Z
Check the source to verify that this is the current version before use.

TABLE I. Nominal resistance value, and maximum resolution, and maximum rated ac or dc working voltage.

| Type designation 1/ | Nominal resistance value (in ohms) 2/ | Maximum resolution 3/ | Maximum rated ac or dc working voltage | Dash number 4/ |
|---|---|---|---|---|
| | Ohms | Percent | Volts | |
| RTR24D-100- | 10 | 1.3 | 2.7 | 001 |
| RTR24D-200- | 20 | 1.1 | 3.8 | 002 |
| RTR24D-500- | 50 | .77 | 6.1 | 003 |
| RTR24D-101- | 100 | .62 | 8.7 | 004 |
| RTR24D-201- | 200 | .55 | 12.3 | 005 |
| RTR24D-501- | 500 | .51 | 19.4 | 006 |
| RTR24D-102- | 1,000 | .37 | 27.4 | 007 |
| RTR24D-202- | 2,000 | .30 | 38.7 | 008 |
| RTR24D-502- | 5,000 5/ | .25 | 61.3 | 009 |
| RTR24D-103- | 10,000 6/ | .19 | 86.7 | 010 |

1/ Complete type designation is for information only and will include additional symbols to complete type designations.
2/ Maximum resolution shown are theoretical.
3/ Maximum resolution shown is theoretical.
4/ Complete dash number will include symbols for terminal type and failure rate level.
5/ Value based on the use of wire having no less than 0.001-inch ±10 percent diameter.
6/ Value based on the use of wire having no less than 0.0008-inch ±10 percent diameter.

3.7 Weight. The maximum weight shall be .00286 pound (1.3 grams).

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-39015.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:30Z
Check the source to verify that this is the current version before use.

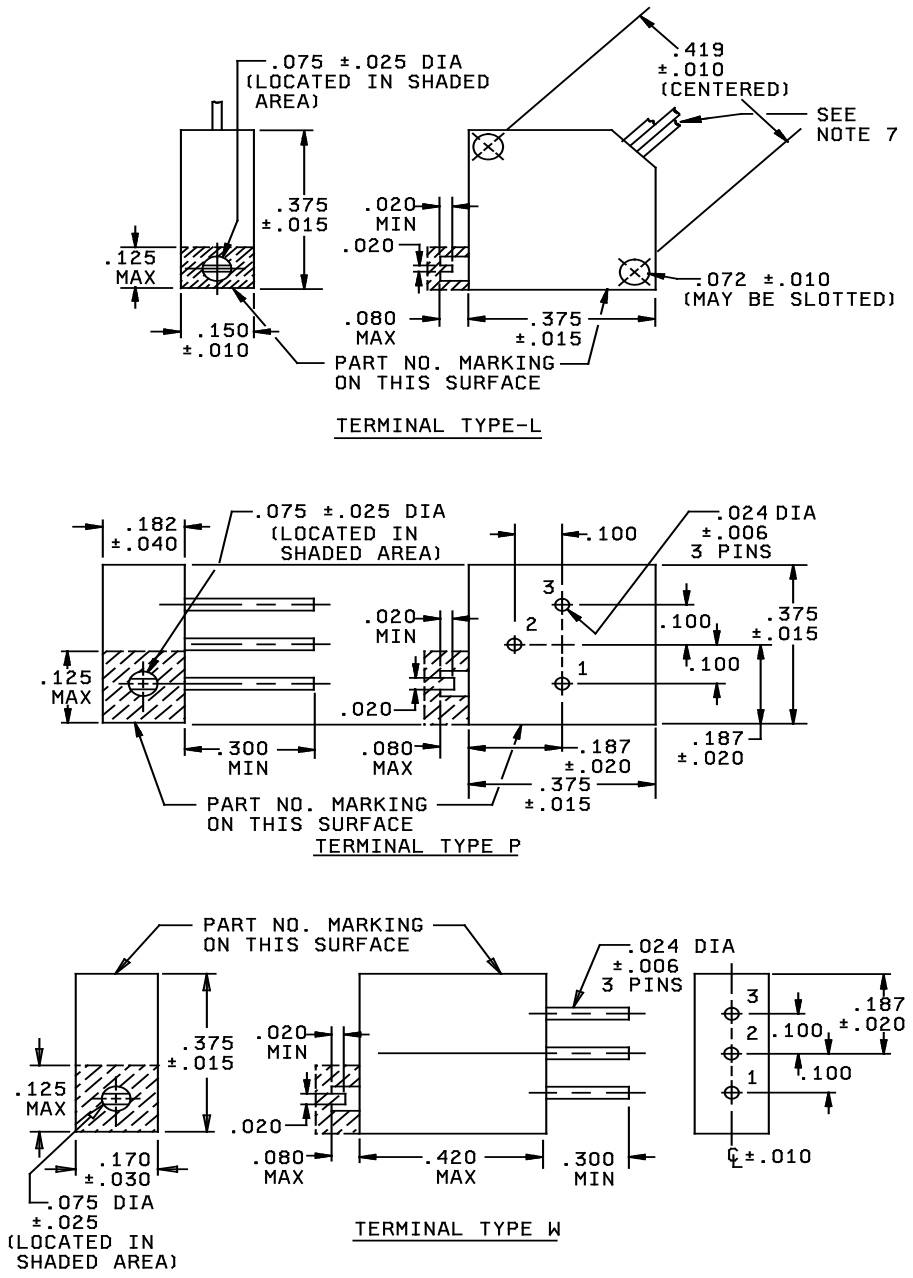
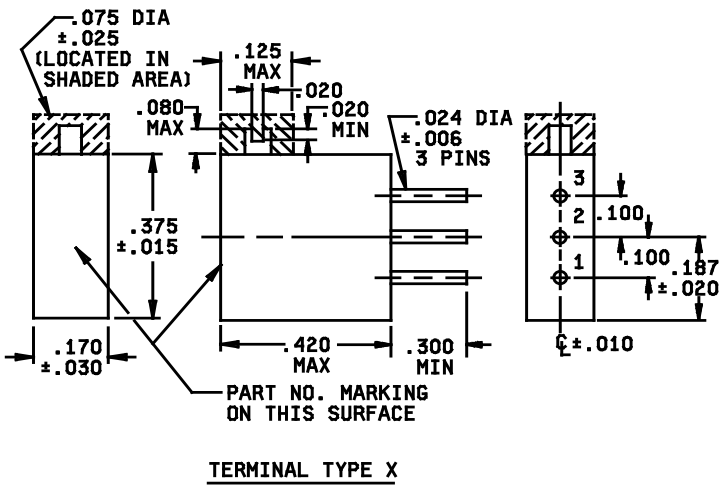



FIGURE 1. Style RTR24 resistors.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:30Z
Check the source to verify that this is the current version before use.




TERMINAL TYPE X

| INCHES | mm |
|---|---|
| .006 | 0.15 |
| .010 | 0.25 |
| .015 | 0.38 |
| .020 | 0.51 |
| .024 | 0.61 |
| .025 | 0.64 |
| .030 | 0.76 |
| .075 | 1.91 |
| .080 | 2.03 |
| .100 | 2.54 |
| .125 | 3.18 |
| .170 | 4.32 |
| .197 | 4.75 |
| .300 | 7.62 |
| .375 | 9.53 |
| .420 | 10.67 |

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.005 (.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information.
4. The entire slot of the actuating screw must be above the surface of the unit.
5. For types P, W, and X, normal mounting means is by use of pins only.
6. The picturization of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
7. The three leads shall be of the stranded wire, AWG size 28 to size 30, having a minimum length of 6 (152.4 mm); they shall be insulated with polytetrafluoroethylene, stripped .250 ±.062 (6.35 mm ±1.57 mm) from the end, and color coded.

FIGURE 1. Style RTR24 resistors - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:30Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-PRF-39015 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date of the specification, and complete PIN.

* b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.

c. Packaging requirements (see 5.1).

6.3 MIL-PRF-27208 substitution data. Resistors of MIL-PRF-39015/3 regardless of their failure rate designation, are substitutes for resistors of the same resistance value, tolerance, terminal characteristic, and resistance temperature specified in the specification MIL-PRF-27208/9.

6.4 Suppression data. Style RTR24 resistors with terminal types P, W, and X is interchangeable with resistors having the same terminal designation under MIL-PRF-39015/3J. The change in dimensions do not require the assignment of a new National Stock Number (NSN) where a NSN had previously been assigned to these items under revisions E, F, G, H, or J of this specification.

* 6.5 Amendment notations. The margins of this specification are marked with asterisks to indicate modification generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5905-2017-050)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:30Z
Check the source to verify that this is the current version before use.